COP2800C Module 6 Graded Programming Assignment

In this assignment, we will use concepts covered in Module 6 to incorporate user-defined methods. Our goal is to break down the program into smaller, reusable methods, improving readability and maintainability while incorporating best practices for method design.

Instructions

1. Accept the GitHub Classroom invite and clone the repo. Using the provided PalmerPenguinsM6.java template file, complete the author and submission date lines in the ID header.
2. Refactor the program to use user-defined methods.
   Do not modify the CSVReader class; all modifications must be made in the PalmerPenguinsM6 class.
   Implement the following methods (**do not remove the TODO markers**)
   Include a descriptive comment above each method definition and call.
   - **TODO 1:** Create a method readSpeciesData that calls CSVReader.readFile(FILE_NAME, 1) and returns the resulting String[]. Replace the inline call to CSVReader.readFile with a call to this method in main.
   - **TODO 2:** Define a method initializeSpeciesCount that returns a new integer array of size NUM_SPECIES. Use this method to initialize speciesCount in main.
   - **TODO 3:** Implement a method isDataEmpty that takes a String[] array as a parameter and returns true if its length is 0, otherwise false. Use this method to check if speciesData is empty in main.
   - **TODO 4:** Write a method countSpecies that accepts String[] speciesData and int[] speciesCount as parameters, iterates through speciesData, and updates speciesCount accordingly. Replace the inline counting logic with a call to this method.
   - **TODO 5:** Implement a method printSpeciesCount that prints the count of each species using the predefined constants. Replace the three System.out.println statements in main with a call to this method.
3. Test Your Program
   - Run your program and verify that it correctly outputs the species counts.
   - Ensure that each method is correctly implemented and called from main.

   Expected output (unchanged from Module 5):

   Chinstrap count = 68
   Gentoo count = 123
   Adelie count = 151

4. **Screenshot Your Output**

- Take a screenshot of your program's output in the "Run I/O" pane of the jGrasp IDE.
- **DO NOT** include the source code in the screenshot.

5. **Submission**
   Submit your completed **PalmerPenguinsM6.java** file to the GitHub repository.
   - Do not include .class files.
   - Ensure your code follows our Java coding conventions (indentation, comments, line length).

   Upload your screenshot to the repository.